

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

How the current solution can scale to handle more users, more data, or extended features.

Current System Limitations:

S.NO	Limitation	Impact	Priority to Address
1	Manual data entry only; no live weather API	Users must know or look up their own readings	High
2	Single-instance deployment (no load balancing configured)	Limited concurrent user capacity under heavy traffic	Medium
3	No automated test suite	Regressions rely on manual verification	Medium

Scalability Plan:

S.NO	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Gunicorn process on Render's free tier	Increase Gunicorn worker count; move to a paid Render plan with autoscaling
2	Data Storage	Static Joblib model files bundled with	Move model artifacts to object storage (e.g., S3) if

		the app	models are updated frequently
3	Performance	Fast (classical ML, sub-second inference)	Add response caching for identical repeated inputs if traffic grows
4	Security	No authentication; public prediction endpoint	Add rate limiting on /predict to prevent abuse if made public at scale

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Live weather API integration for automatic input	+1-2 months	Removes manual data entry friction, broadens usability
Phase 3	River/dam water-level data as additional model features	+3-4 months	Improves prediction accuracy with more direct hydrological signals
Phase 4	Time-series forecasting for multi-day lead time	+6 months	Moves from a snapshot classifier to a forward-looking forecast